

ARYAN BAHL

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EDUCATION

University of Illinois at Urbana-Champaign

Master's of Computer Science

Expected: December 2026

B.S. in Computer Science | Minor in Statistics

May 2025

PROFESSIONAL EXPERIENCE

Member of Technical Staff

March 2026 – Present

Endeavor

- Own end-to-end production integrations for five enterprise clients representing \$400k in contract value, delivering PO extraction, matching, and ERP write-back
- Improve product-matching accuracy by 15% across 400+ live order documents by enriching Pinecone vector retrieval with extracted customer item codes and catalog metadata
- Cut catalog-upload infrastructure cost by 90% and eliminated a 20% job-failure rate by re-architecting the distributed pipeline from an always-on ECS worker to an event-driven AWS Step Functions orchestrator with per-phase retries
- Design an agentic ingestion runtime for a multi-tenant ontology platform using LangGraph StateGraph wrapping Claude via DeepAgents, with a Redis checkpoint for mid-run resume and LangSmith tracing for per-run observability

Head of Engineering

June 2025 – March 2026

Nora Music

- Led engineering for the platform end-to-end, designing a horizontally-scalable architecture that addressed primary-database bottlenecks through Redis and read replicas
- Cut third-party data ingestion time by 75% by parallelizing stat collection pipelines using AWS SQS and Glue jobs
- Identified app performance issues by analyzing PostHog metrics, resulting in a 400% increase in average session length

Machine Learning Engineer

March 2025 – January 2026

Boston Bioprocess

- Halved engineer workload by building an automated experiment summarization system that triggered on upload and streamed LLM-generated summaries directly to users
- Lowered cloud infrastructure costs by 18% through optimized GPU utilization during ML training pipelines on AWS SageMaker

PROJECTS

Lead Enrichment Workflow | *Celery, Redis, FastAPI, Next.js*

- Built a block-based workflow tool to replace manual data scraping, enabling users to configure pipelines with run history tracking
- Improved reliability under third-party APIs by using Celery and Redis with idempotent task guarantees and retry semantics

Public Speaking Assistant | *React, Django, TensorFlow, MongoDB, S3*

- Built a full-stack web platform that allowed users to privately upload speech recordings and receive structured feedback across clarity, enthusiasm, engagement, and assertiveness
- Reduced audio analysis latency by parallelizing ML classification and offloading file storage to AWS S3

LEADERSHIP

Outreach Chair

May 2025 – February 2026

HackIllinois

- Raised \$110k+ in sponsorship funding by reimagining outreach strategy to prioritize warm leads and measurable sponsor ROI
- Coordinated logistics for a 1,000+ person hackathon that doubled attendee engagement and project submissions from previous years

Systems Development Chair

January 2025 – September 2025

Reflections | Projections

- Directed a 10-person engineering team to design and deploy production infrastructure supporting 3,000+ concurrent event attendees
- Standardized development environments by building a Docker-based full-stack container complete with a local Supabase instance

SKILLS

Languages: Python, C++, Java/Kotlin, TypeScript, SQL

Frontend: React, React Native, Next.js, Tailwind

Backend: FastAPI, Node.js, Redis, PostgreSQL, Celery, REST APIs, microservices

Cloud/Infra: AWS (Lambda, ECS, EC2, Aurora, CloudWatch, SageMaker, Step Functions), Docker, Terraform

Concepts: Distributed systems, system design, horizontal scaling, observability/monitoring, caching